AI-Powered Smart Water Management Platform

Complete Project Development Context

Core focus: Intelligent Water Demand Forecasting, Water Stress Analysis and Smart Supply Planning.

Current scope: Population + Population Density + Historical Consumption + Reservoir + Time/Season → AI Forecasting → Water Availability → Water Stress → Smart Supply Plan.

1. Project Overview

The proposed project is an AI-powered municipal water-management platform designed to help authorities predict future water demand and plan water distribution more effectively. The system uses historical water consumption together with population, population density, reservoir availability and temporal/seasonal patterns to forecast future water requirements.

Instead of depending only on fixed water-supply schedules or manual decisions, the system answers: **How much water will each area need in the coming days, and will the available reservoir water be sufficient to meet that demand?**

Multiple AI/ML models—XGBoost, LSTM, GRU and Transformer—will be evaluated. The best-performing model will be selected using forecasting metrics. Its demand forecasts will then be compared with reservoir availability to identify potential water-stress or shortage conditions, followed by a smart supply recommendation.

2. Actual Problem Being Solved

Municipal water demand is not constant. Different wards have different population sizes and population densities, so their water requirements are different. Fixed or manually prepared schedules may therefore cause under-supply in some areas and over-supply in others.

Municipalities may already have historical consumption, population and reservoir records, but these datasets are not necessarily integrated for predictive planning. Existing research often treats demand forecasting and water-level/reservoir forecasting as separate problems.

The project aims to move from fixed/manual planning toward predictive, data-driven and resource-aware water-supply planning.

3. Core Research Question

> **Can AI use population, historical water consumption, reservoir availability and temporal patterns to predict future water demand and help municipalities make better water-supply decisions?**

4. Core Input Data

Population Data

Ward ID, ward name, year, population, area, population density, population growth.

Historical Water Consumption

Date, Ward ID, water consumption, water supplied, daily/weekly/monthly demand.

Reservoir Data

Reservoir ID, date, water level, storage capacity, current storage, storage percentage, available water.

Time / Seasonal Data

Date, day, day of week, month, year, season, weekend.

Scope decision: Rainfall/weather prediction is not a primary input in the current architecture. The core system is based on population, consumption, reservoir and temporal information.

5. Complete System Workflow

```
Population + Population Density
+
Historical Water Consumption
+
Reservoir Data
+
Time / Season
↓
Data Preprocessing
↓
Feature Engineering
↓
Dataset Creation
↓
Model Training
↓
XGBoost / LSTM / GRU / Transformer
↓
Model Evaluation
↓
Best Model Selection
↓
Water Demand Forecast
↓
■■■■■■■■■■■■■■■■■■■■
↓ ↓
Reservoir Demand–Supply
Availability Analysis
↓ ↓
■■■■■■■■■■■■■■■■■■■■
↓
Water Stress
Analysis
↓
Smart Supply Plan
↓
Admin / GIS / Citizen App
```

6. Data Collection Module

Collect population, historical consumption, reservoir and temporal records and link them through common time and geographical identifiers. A key such as Ward_ID + Date can be used where the actual datasets support ward-level daily records.

7. Data Preprocessing Module

• Remove duplicate records.

• Handle missing values.

• Detect abnormal values/outliers.

• Standardise dates and ward/zone IDs.

• Merge datasets using time and location keys.

• Scale/normalise features where required.

The output is a clean integrated dataset ready for feature engineering and modelling.

8. Feature Engineering

Population Features

• Population

• Population density

• Population growth

Consumption Features

• Previous-day demand

• Previous-7-day demand

• 7-day rolling average

• 30-day rolling average

• Previous-month demand

Reservoir Features

• Water level

• Storage

• Storage percentage

• Available water

Temporal Features

• Month

• Day

• Day of week

• Year

• Season

• Weekend

9. Dataset Creation

For time-series forecasting, split data chronologically so future records are not mixed into training. A possible starting split is 70% training, 15% validation and 15% testing; the final split should depend on dataset size.

10. AI Forecasting Module

XGBoost

Tree-based ensemble model useful for tabular features such as population, density, lag demand, reservoir variables and seasonal features.

LSTM

Recurrent neural network designed for sequential data and temporal consumption patterns.

GRU

Simpler recurrent architecture that provides another sequential forecasting baseline.

Transformer

Attention-based architecture useful for long sequences and multi-step forecasting.

All four models should be evaluated under comparable conditions rather than assuming that the most complex model will perform best.

11. Model Evaluation and Selection

Model	MAE	RMSE	MAPE	R^2
XGBoost	—	—	—	—
LSTM	—	—	—	—
GRU	—	—	—	—
Transformer	—	—	—	—

MAE: average absolute error. **RMSE:** emphasises larger errors. **MAPE:** percentage error. **R^2:** explained variation.

12. Multi-Horizon Forecasting

Generate 1-day, 3-day and 7-day demand forecasts to support immediate, short-term and weekly planning.

13. Reservoir Availability Analysis

Analyse water level, storage capacity, current storage and available storage.

```
Storage Percentage = (Current Storage / Total Capacity) × 100
```

This provides the resource side of the demand–availability problem.

14. Demand–Supply Analysis

Compare predicted demand with available reservoir water. Example:

```
Forecasted Demand = 500 ML
Available Water = 450 ML
Potential Gap = 50 ML
```

This gap becomes an input to the water-stress decision layer.

15. Water Stress Analysis

A Water Availability Ratio (WAR) can be used as a decision-support indicator:

```
WAR = Available Water / Forecasted Demand
```

For example, 450 ML available against 500 ML forecast demand gives WAR = 0.90. Final stress categories and thresholds should be calibrated to the selected municipal context.

- Safe
- Moderate
- High Stress
- Shortage

16. Smart Supply Planning

Use predicted demand, reservoir availability, water stress and zone-wise requirements to generate supply recommendations. Linear Programming or MILP can be introduced for advanced allocation optimisation.

Conceptual objective: minimise supply deficit while respecting available water and zone-level requirements.

17. GIS Module

Visualise ward boundaries, population density, predicted demand, water-stress zones, supply status and reservoir locations.

Technology: Leaflet.js + OpenStreetMap.

18. Admin Dashboard

- View demand forecasts.
- Monitor population and reservoir status.
- View water-stress zones.
- See smart supply recommendations.
- View GIS maps.
- Review leakage/wastage indicators.
- Manage complaints and reports.

Frontend: React.js. Charts can use Recharts or Plotly.

19. Citizen Application

• View water-supply schedule.

• Receive shortage alerts.

• Receive maintenance/service notifications.

• Register complaints.

• Track complaint status.

Technology: Flutter.

20. Leakage and Wastage — Supporting Modules

Leakage prediction remains a supporting module because it is part of the broader project scope, but the main research focus is demand forecasting, reservoir-aware water-stress analysis and supply planning.

```
Water Loss = Water Supplied – Water Consumed
```

Random Forest/XGBoost can be used for leakage-risk classification when suitable historical labels exist. Isolation Forest can flag unusual supply/consumption patterns.

21. Backend and API Layer

FastAPI + Python connects AI models with the web dashboard, mobile app and database.

```
React Dashboard
■
Flutter App
■
↓
FastAPI
■
■■■■■■■■■■■■■■■■■
↓ ↓ ↓
ML PostgreSQL GIS
```

Possible APIs: /api/population, /api/consumption, /api/reservoir, /api/forecast, /api/water-stress, /api/supply-plan, /api/leakage-risk and /api/complaints.

22. Database Design

Database: PostgreSQL

• Users

• Wards

• Population

• Consumption

• Reservoir

• Forecast

• WaterStress

• SupplyPlan

• Complaints

• Maintenance

Core relationships: Ward → Population/Consumption/Forecast/WaterStress/SupplyPlan; Reservoir → Availability → Water Stress; Forecast + Availability → Smart Supply Plan.

23. Object-Based Design

• User → Admin / Citizen

• Ward

• Population

• Consumption

• Reservoir

• Forecast

• WaterStress

• SupplyPlan

• Complaint

• MaintenanceRecord

Important relationships: Ward → Consumption; Ward → Forecast; Reservoir → Availability; Forecast + Availability → Water Stress; Water Stress + Demand → Supply Plan.

24. Security

• JWT authentication

• Role-based access control

• Password hashing

• Authenticated APIs

25. Recommended Technology Stack

Layer	Technology
Programming	Python, JavaScript/TypeScript, Dart
Data Processing	Pandas, NumPy
ML	Scikit-learn, XGBoost
Deep Learning	TensorFlow / Keras
Forecasting	XGBoost, LSTM, GRU, Transformer
Evaluation	MAE, RMSE, MAPE, R^2
Database	PostgreSQL
Backend	FastAPI
Admin Dashboard	React.js
Mobile App	Flutter
GIS	Leaflet.js + OpenStreetMap
Charts	Recharts / Plotly
Authentication	JWT
Notifications	Firebase Cloud Messaging
Deployment	Render + Netlify

26. Complete Technical Architecture

```
DATA LAYER
Population | Consumption | Reservoir | Time/Season
↓
DATA PROCESSING
Cleaning | Integration | Feature Engineering
↓
AI/ML LAYER
XGBoost | LSTM | GRU | Transformer
↓
```

```
MODEL EVALUATION
MAE | RMSE | MAPE | R²
↓
WATER DEMAND FORECAST
1-Day | 3-Day | 7-Day
↓
■■■■■■■■■■■■■■■■■■■■■■■■■■■■
↓ ↓
Reservoir Availability Demand–Supply
↓ ↓
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↓
Water Stress
↓
Smart Supply Plan
↓
Admin | GIS | Citizen
```

27. Proposed Research Contribution

• **Population-aware forecasting:** include population and population density in demand modelling.

• **Reservoir-aware planning:** consider available water alongside predicted demand.

• **Multi-model comparison:** evaluate XGBoost, LSTM, GRU and Transformer.

• **Multi-horizon forecasting:** 1-day, 3-day and 7-day predictions.

• **Demand–availability analysis:** explicitly compare predicted demand with available resources.

• **Water-stress identification:** use a calibrated availability-based indicator.

• **Smart supply recommendation:** convert forecast results into planning recommendations.

• **Municipal decision support:** integrate results into dashboard/GIS/mobile interfaces.

28. Development Roadmap

Phase 1 — Dataset — Find Indian/municipal population, consumption and reservoir data; identify common time/location keys.

Phase 2 — Data Engineering — Clean, merge and validate datasets; create target and features.

Phase 3 — ML Prototype — Build XGBoost, LSTM, GRU and Transformer baselines.

Phase 4 — Evaluation — Compare MAE, RMSE, MAPE and R^2; select the best model.

Phase 5 — Forecasting API — Integrate the selected model with FastAPI.

Phase 6 — Water Stress — Implement demand–availability gap and calibrated stress logic.

Phase 7 — Smart Planning — Develop zone-wise supply recommendation/optimisation.

Phase 8 — Dashboard — Build React municipal dashboard.

Phase 9 — GIS — Add Leaflet/OpenStreetMap visualisation.

Phase 10 — Citizen App — Build Flutter application and notifications.

Phase 11 — Supporting Modules — Add leakage risk, wastage/anomaly analysis, complaints and alerts.

29. Recommended Development Priority

Do not build every module simultaneously. The first working prototype should prove the research core:

```
Dataset → Preprocessing → Feature Engineering
→ AI Forecasting → Demand Forecast
→ Reservoir Availability → Water Stress
```

Once this works, add the smart supply planner, dashboard, GIS and citizen application. Leakage, wastage, complaints and notifications can then be added as supporting modules.

30. Example End-to-End Output

```
Input:
Ward = A
Population = 50,000
Density = High
Historical Demand = 480 ML
Reservoir Availability = 450 ML
Season = Summer

↓

Predicted Demand = [Model Output]
Available Water = 450 ML
Demand–Supply Gap = [Calculated]
Water Stress = [Calculated]
Supply Recommendation = [Calculated]
```

31. Final Project Definition

“We are developing an AI-powered municipal water-management platform that forecasts future water demand using population, population density, historical consumption, reservoir and temporal data. We compare the predicted demand with available reservoir resources to identify potential water-stress conditions and generate intelligent zone-wise water-supply recommendations. Multiple models such as XGBoost, LSTM, GRU and Transformer will be experimentally compared, and the best-performing model will be integrated into a municipal decision-support platform with dashboard and GIS visualisation.”

32. One-Line Project Concept

Population + Consumption + Reservoir + Time → AI Forecast → Water Availability → Water Stress → Smart Supply Plan